EXPERIMENT 5

Weather Dashboard using React Hooks and OpenWeatherMap API

AIM

To build a Weather Dashboard using React Hooks that fetches and displays weather information from the OpenWeatherMap API.

ALGORITHM

1. Create a React functional component.
2. Use useState to store city, weather and error values.
3. Fetch weather data when the user searches for a city.
4. Update state with the API response.
5. Display temperature, weather description and humidity.

PROGRAM

```
// App.jsx
import {useState} from 'react';

export default function App(){
 const [city,setCity]=useState('');
 const [weather,setWeather]=useState(null);
 const [error,setError]=useState('');
 const API_KEY='YOUR_API_KEY';

 async function searchWeather(){
  try{
   setError('');
   const res=await fetch(
    `https://api.openweathermap.org/data/2.5/weather?q=${city}&appid=${API_KEY}&units=metric`);
   if(!res.ok) throw new Error('City not found');
   setWeather(await res.json());
  }catch(e){setWeather(null);setError(e.message);}
 }
 return <div>
  <h2>Weather Dashboard</h2>
  <input value={city} onChange={e=>setCity(e.target.value)} placeholder="City"/>
  <button onClick={searchWeather}>Search</button>
  <p>{error}</p>
  {weather && <div>
   <h3>{weather.name}</h3>
   <p>{weather.main.temp} °C</p>
   <p>{weather.weather[0].description}</p>
   <p>Humidity: {weather.main.humidity}%</p>
  </div>}
 </div>;
}
```

RESULT

Thus, a weather dashboard was developed successfully using React Hooks and the OpenWeatherMap API.

VIVA QUESTIONS

1. What are React Hooks?
2. What is the use of useState()?
3. What is an API?
4. Why is fetch() used?
5. What information does the weather API return?